

***Global Cancer Patient Analysis & Prediction System
(Classification)***

Course: Big Data Analytics

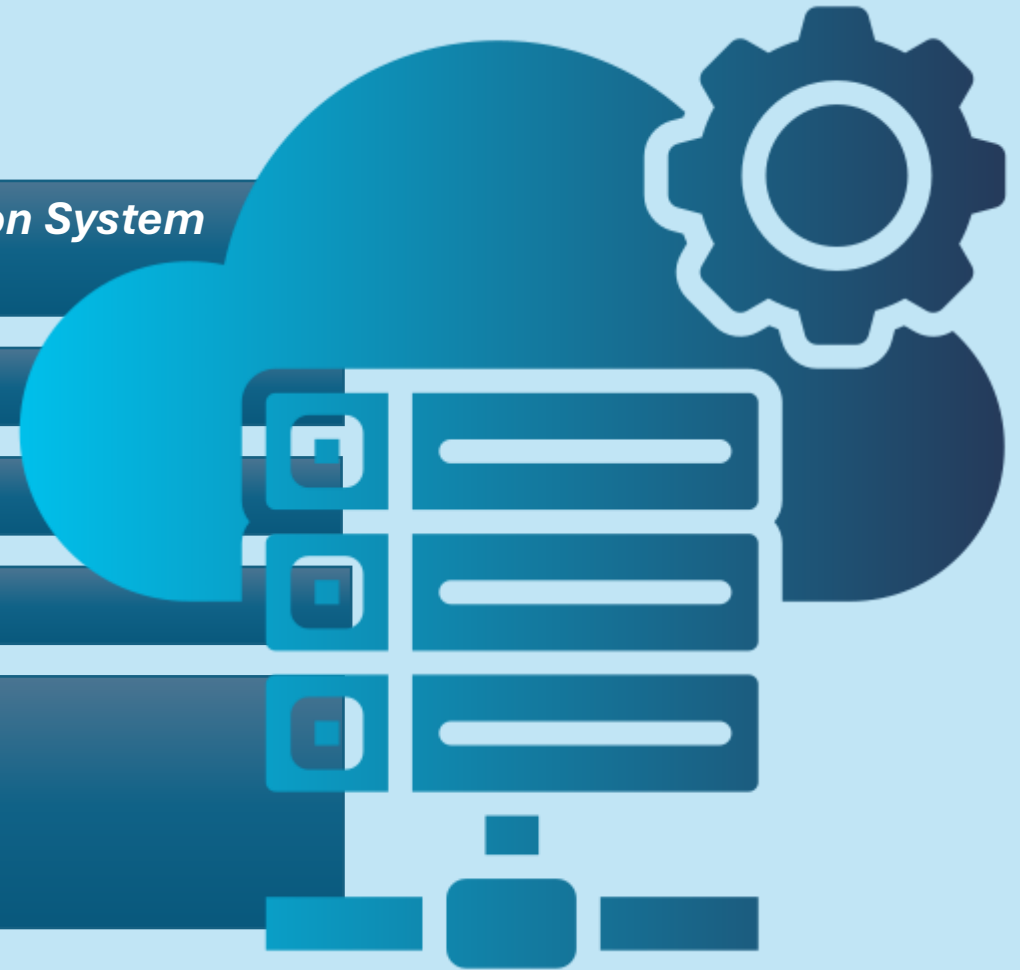
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Dataset Analysis (Overview)

- ***Dataset Name: global_cancer_patients_2015_2024.csv***
- ***Total Records (Rows): 50,000***
- ***patients Total Attributes (Columns): 15 features***
- ***Data Integrity: Complete dataset with 0 missing values.***
- ***Attribute Types:***
 - ***Categorical: Gender, Country, Cancer Type, Cancer Stage.***
 - ***Numerical: Age, Year, Treatment Cost, Survival Years.***
 - ***Risk Factors (Scales 0-10): Genetic Risk, Smoking, Alcohol Use, Obesity, Air Pollution.***



Dataset Preview

- **Observations:**

- Each row represents a unique patient (PT000...).
- Contains a mix of text (Cancer Type) and numerical scores (Genetic Risk).
- Target Variable: Target_Severity_Score (calculated metric).



Data Preprocessing

- **1. Data Structure Analysis**
- **Dimensions:** The dataset consists of **50,000 rows** and **15 columns**.
- **Data Types:**
 - **Qualitative (Object):** Patient_ID, Gender, Country_Region, Cancer_Type, Cancer_Stage.
 - **Quantitative (Int/Float):** Age, Year, Treatment_Cost_USD, Survival_Years.
 - **Risk Scores (Float):** Genetic_Risk, Air_Pollution, Smoking, etc.
- **2. Duplicate Detection**
- **Method:** Checked for identical rows using the duplicated() function.
- **Result: 0 Duplicates found.** The data is unique for every patient record.
- **3. Missing Value Analysis (NA)**
- **Method:** Scanned all columns for null or NA values.
- **Result: 0 Missing values.**
 - *Insight:* The dataset is complete and does not require imputation (filling in blanks), which increases the reliability of our model.
- **4. Outlier Detection & Handling**
- **Method:** Used the **Interquartile Range (IQR)** method to detect extreme values.
- **Findings:**
 - Most columns (Age, Cost, Genetic Risk) showed a normal distribution with **no significant outliers**.
 - **Target_Severity_Score:** Detected **151 outliers**.

Algorithm Explanation: Multinomial Logistic Regression

1. Why did we choose this specific algorithm?

Standard "Binary Logistic Regression" is used when there are only two possible outcomes (e.g., Sick vs. Healthy).

However, in your dataset, we grouped the `Target_Severity_Score` into **3 distinct classes**:

1.Low

2.Medium

3.High

Since we have **more than two categories** to predict, **Multinomial Logistic Regression** is the correct mathematical approach because it handles multi-class classification problems effectively.

2. How does it work on your data?

The algorithm calculates the **probability** that a specific patient belongs to each of the three categories (Low, Medium, High) based on their input features.

•**Inputs (Features - X):** The risk factors from your dataset:

- `Genetic_Risk`
- `Smoking`
- `Alcohol_Use`
- `Obesity_Level`
- `Air_Pollution`

•**Output (Target - Y):** The predicted `Severity_Class`.

- **3. The Logic (Simplified)**

- The model builds linear equations for each class relative to a "reference" class (e.g., comparing "High" vs. "Low").

- **4. Practical Example from Your Dataset**

- Imagine a patient with the following data:

- **Smoking:** 9.0 (Very High)

- **Genetic_Risk:** 8.5 (High)

- **What does the algorithm do?**

- **Calculates Probabilities:**

- Probability of **Low**: 5%
- Probability of **Medium**: 20%
- Probability of **High**: 75%

- **Final Decision:** Since 75% is the highest probability, the model classifies this patient as "**High Severity**."



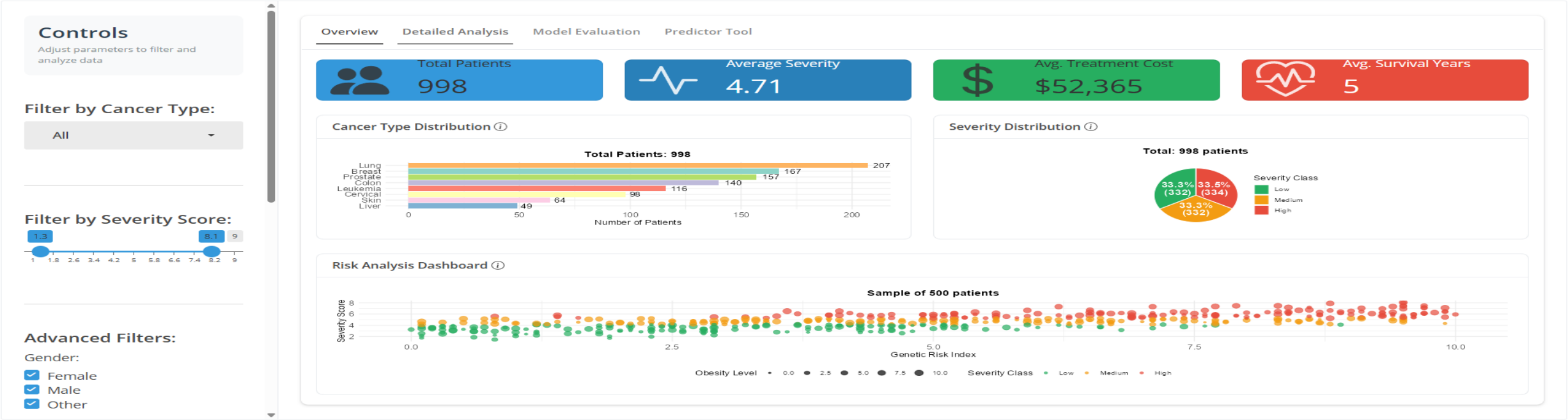
The Final Product (Shiny GUI)

□ Interface Features:

- **Left Sidebar:** Filters for Dataset (Age, Country, Cancer Type).
- **Predictor Tool:** Sliders for real-time input of patient risk factors.
- **Output:** Instant classification (Severity Level) and probability charts.
- **User Utility:** Designed for doctors to quickly assess patient risk profiles.

Cancer Analysis Dashboard

Comprehensive analysis and prediction system for cancer patients



Key Functions:

1.Interactive Data Exploration:

- Allows users to **filter** the entire dataset by Age, Gender, Country, and Cancer Type via the sidebar control panel.
- Automatically updates visualizations (charts and graphs) based on the selected filters to show specific trends.

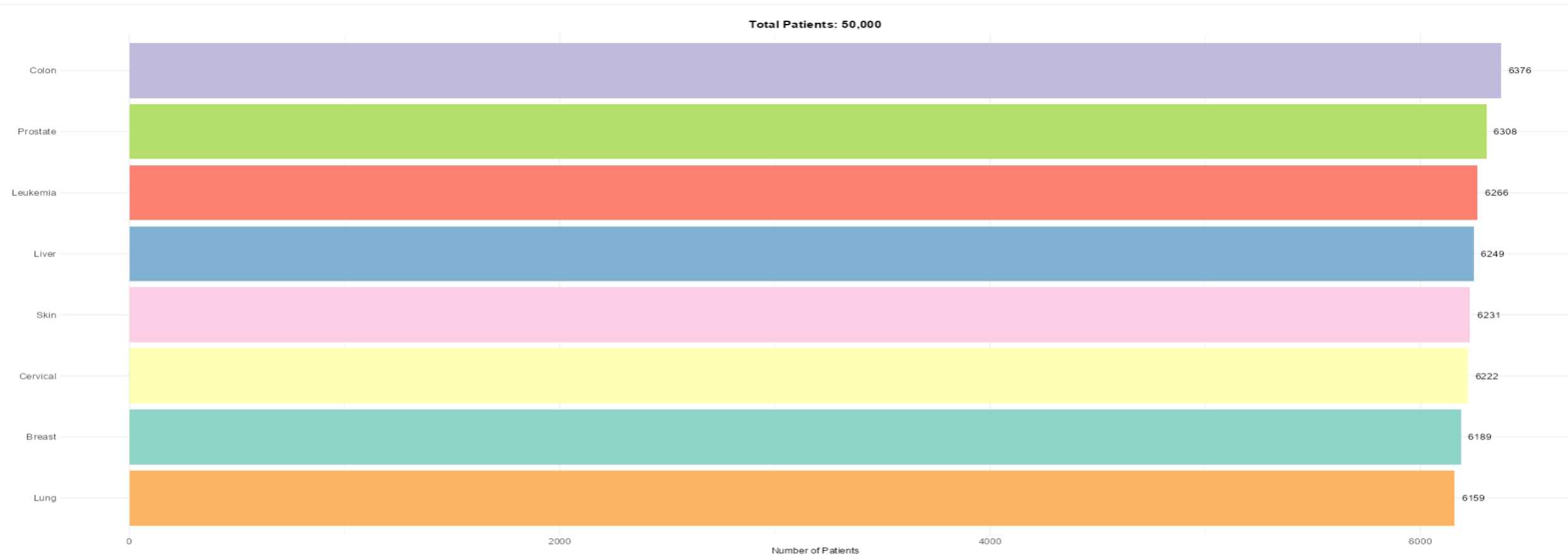
2.Predictive Diagnostics (The "Predictor Tool"):

- Provides a user-friendly interface where doctors can input a new patient's risk factors (e.g., Genetic Risk, Smoking, Obesity) using simple **sliders**.
- Uses the background Machine Learning model to instantly calculate and display the **Predicted Severity Class** (Low, Medium, or High) along with the confidence probability.

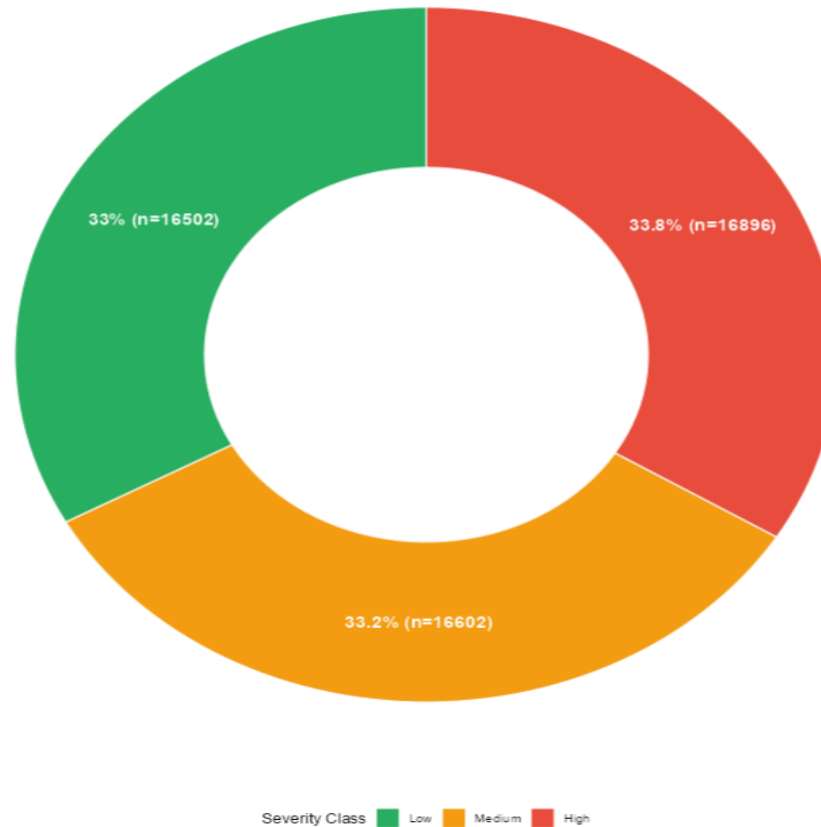
3.Model Evaluation & Transparency:

- Displays the performance metrics of the algorithm (Accuracy, Confusion Matrix) to build trust in the system's predictions.
- Includes a detailed **Data Table** that allows researchers to view raw records and export them to Excel/CSV for further analysis.

Cancer Type Distribution ⓘ



•**Bar Plot:** A horizontal bar chart is the standard way to compare quantities (number of deaths) across different categories (cancer types).



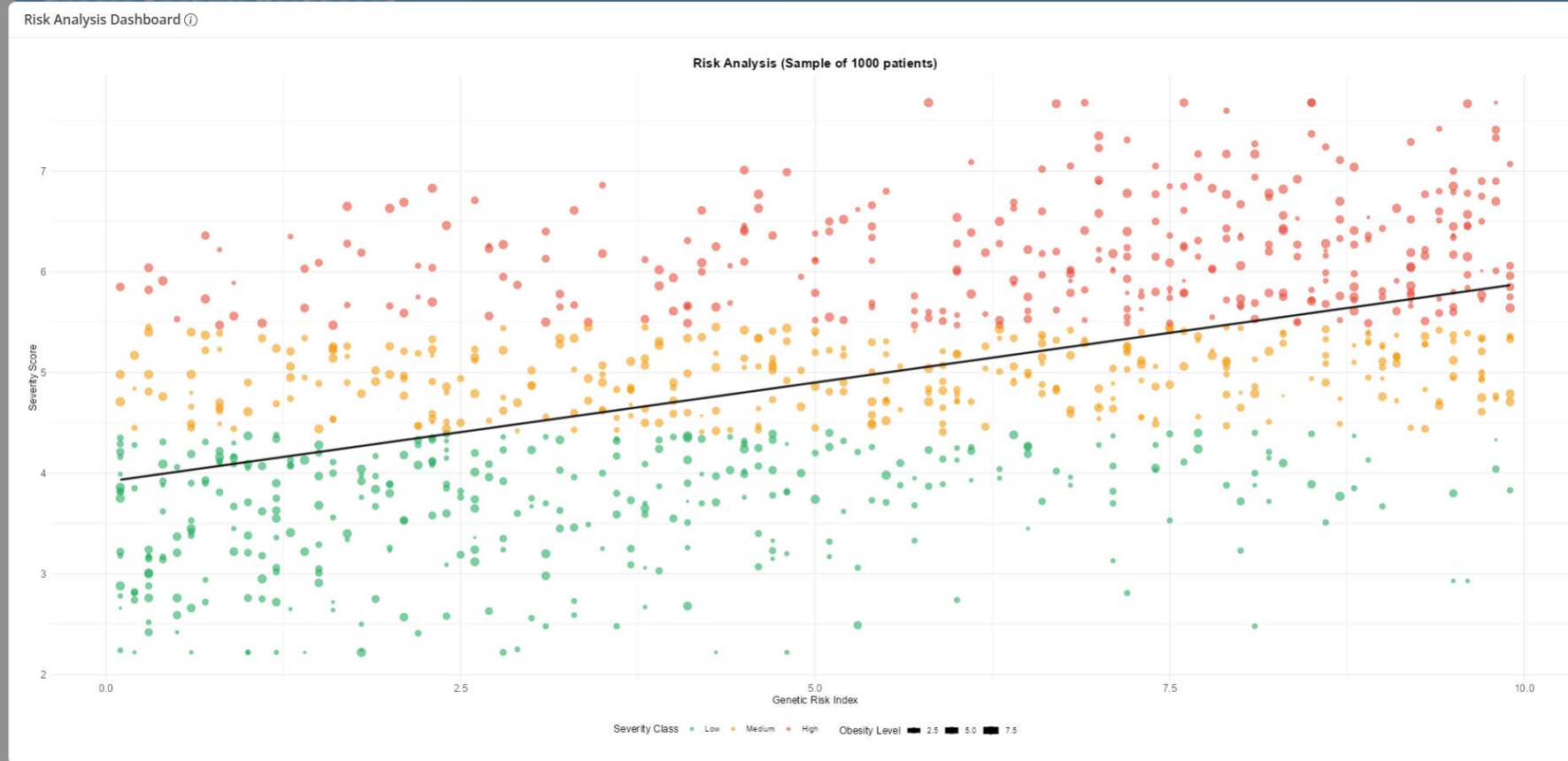
Overview For Pie Chart

The chart visualizes the classification of **998 patients** based on the severity of their condition. The data is categorized into three distinct classes: **Low**, **Medium**, and **High**.

Data Breakdown

The distribution is remarkably even across all three categories:

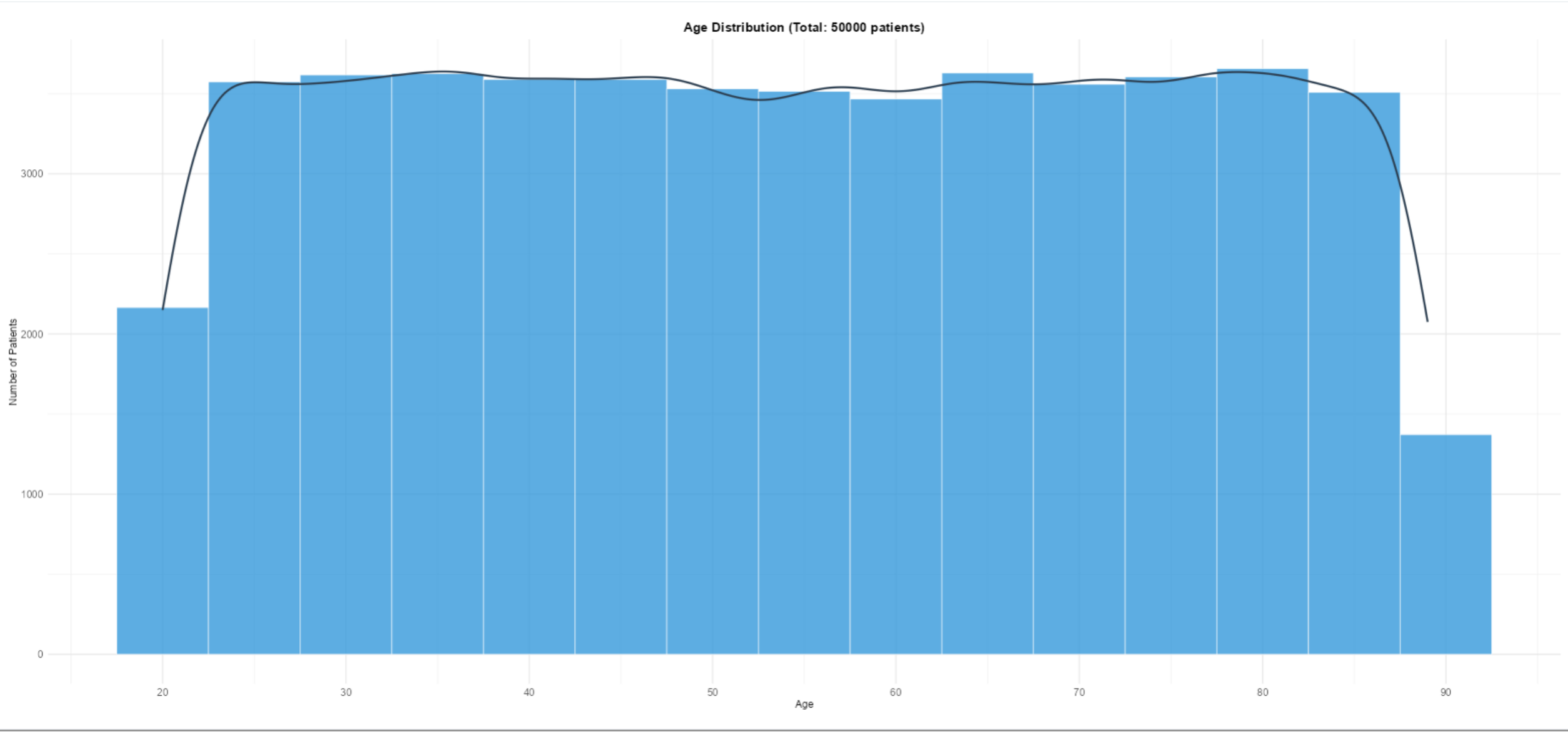
- **High Severity (Red Slice):** Represents **33.5%** of the population, which accounts for **334** patients.
- **Medium Severity (Orange Slice):** Represents **33.3%** of the population, which accounts for **332** patients.
- **Low Severity (Green Slice):** Represents **33.3%** of the population, which accounts for **332** patients.



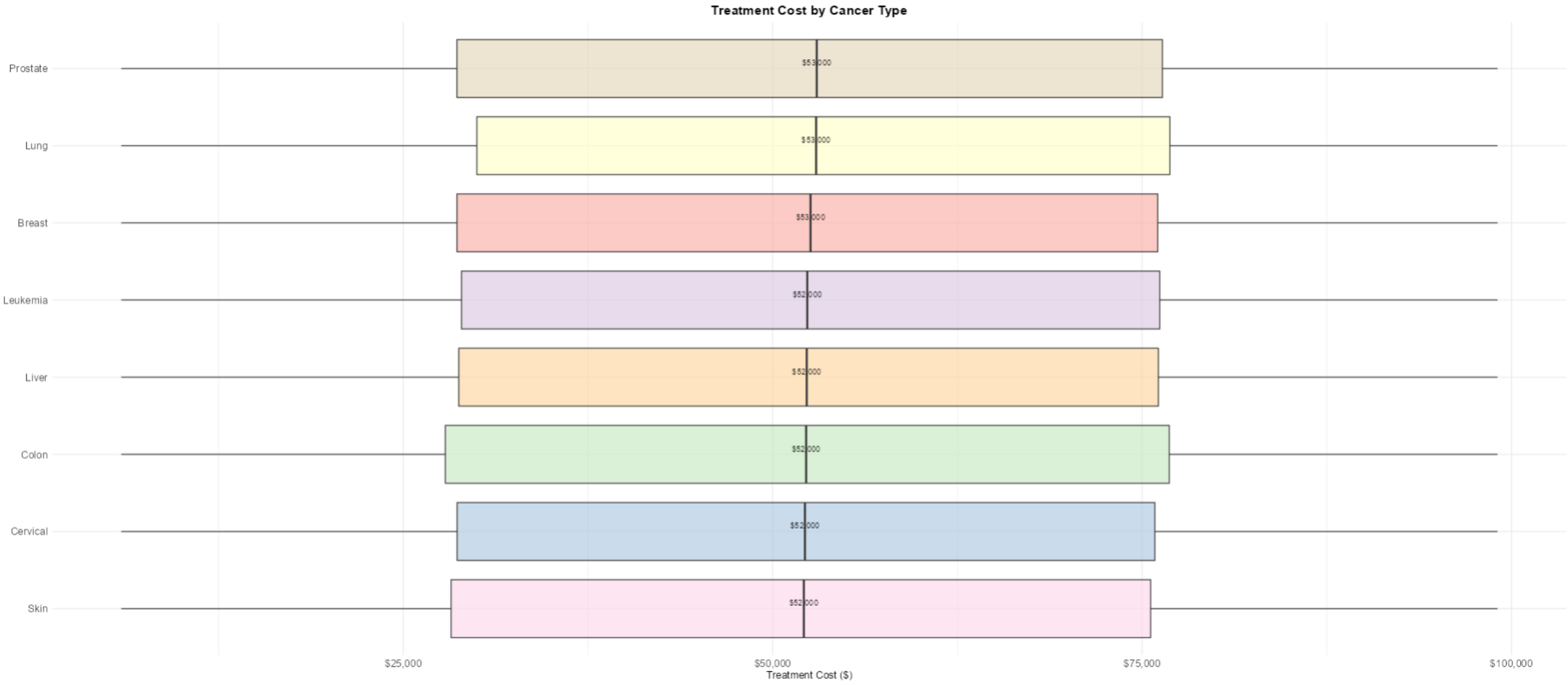
Scatter Plot

Advanced Version

- ❑ This is **a bubble chart** showing a positive correlation where higher Genetic Risk leads to a higher Severity Score. The points are colored by severity class and sized by Obesity Level, indicating that patients with higher genetic risk and obesity tend to have the most severe conditions.



☐ **Age Demographics (Histogram):** The histogram displays a fairly uniform distribution of **998 patients** across ages 18 to 92, indicating the dataset is balanced and includes a wide variety of age groups.



❑ Treatment Cost Analysis: The box plots reveal that **Skin cancer** treatment has the highest median cost and variability, while **Leukemia** is the least expensive.

Cancer Analysis Dashboard

Comprehensive analysis and prediction system for cancer patients

Controls

Adjust parameters to filter and analyze data

Filter by Cancer Type:

All

Filter by Severity Score:



Advanced Filters:

Gender:

- ☒ Female
- ☒ Male
- ☒ Other

Overview Detailed Analysis Model Evaluation Predictor Tool

Patient Diagnostics

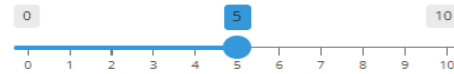
Run Diagnostics

Input Parameters ⓘ

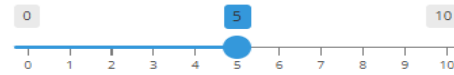
Genetic Risk



Air Pollution



Alcohol Use



Smoking



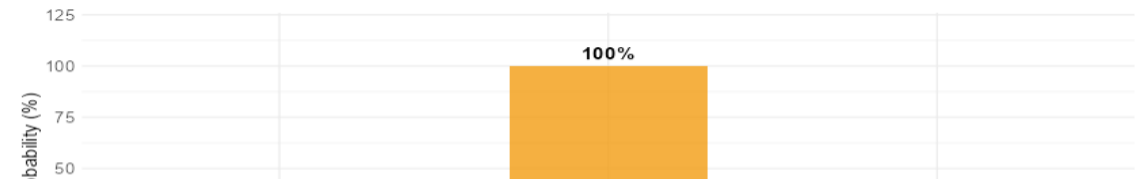
Diagnostic Results

Estimated Severity Classification:

Medium

Severity Level

Confidence Probability Breakdown:



- ❑ displays a **Cancer Analysis Dashboard** interface featuring a "Predictor Tool" that allows users to input patient risk factors (like genetic risk and lifestyle habits) using adjustable sliders.
- ❑ Based on these inputs, the system calculates and displays a diagnostic result, shown here as an estimated **"Medium" severity classification** with a corresponding confidence probability chart

1. Algorithm: Logistic Regression

Type: Classification Algorithms

```
final_model <- multinom(Severity_Class ~ Genetic_Risk + Air_Pollution +  
                        Alcohol_Use + Smoking + Obesity_Level,  
                        data = trainData, trace = FALSE)
```

❑ Model Characteristics:

•**Number of Classes:** 3 classes (Low, Medium, High)

•**Independent Variables (Features):**

- Genetic_Risk
- Air_Pollution
- Alcohol_Use
- Smoking
- Obesity_Level

•**Dependent Variable (Target):** Severity_Class

❑ Why This Algorithm Was Chosen:

Advantages:

- Interpretability:** Easy to understand feature importance
- Probabilistic Output:** Provides probabilities, not just classifications
- Fast Training:** Efficient for moderate-sized datasets
- No Feature Scaling Needed:** Works well with preprocessed data
- Handles Multiclass Naturally:** Built for >2 classes

Limitations:

- Assumes Linear Relationships:** May not capture complex patterns
- Can Overfit** with many features
- Requires Feature Independence** assumption

2. XGBoost (eXtreme Gradient Boosting) Algorithm

What It Is:

XGBoost is an advanced ensemble learning algorithm that builds multiple decision trees sequentially, with each tree correcting the errors of the previous one. It's particularly effective for structured/tabular data.

Key Characteristics :

- Sequential Learning:** Trees are built one after another
- Gradient Boosting:** Uses gradient descent to minimize errors
- Regularization:** Built-in to prevent overfitting
- Handles Missing Values:** Automatically
- Feature Importance:** Can rank features by contribution

```
r                                                                    Copy Download

# XGBoost Configuration
params <- list(
  objective = "multi:softprob", # Multi-class classification
  num_class = 3,                # 3 severity classes (Low, Medium, High)
  eta = 0.1,                    # Learning rate (step size)
  max_depth = 6,                # Maximum tree depth
  eval_metric = "mlogloss"      # Evaluation metric
)
```

Advantages :

- High Accuracy:** Often achieves superior performance
- Speed:** Optimized for performance
- Flexibility:** Handles various data types
- Robustness:** Less prone to overfitting than traditional methods

3. Random Forest Algorithm

What It Is:

Random Forest is an ensemble learning method that builds multiple decision trees independently and combines their predictions through majority voting (for classification).

```
r                                                                    Copy Download

# Random Forest Configuration
rf_model <- randomForest(Severity_Class ~ Age + Genetic_Risk + ...,
                        data = trainData,
                        ntree = 100,      # Number of trees
                        importance = TRUE) # Calculate feature importance
```

Advantages :

Key Characteristics :

- **Parallel Learning:** All trees built independently
- **Bagging (Bootstrap Aggregating):** Each tree trained on random data subset
- **Feature Randomness:** Each split considers random feature subset
- **Majority Voting:** Final prediction = most frequent class prediction

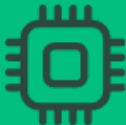
- **Reduced Overfitting:** Due to ensemble averaging
- **Feature Importance:** Built-in calculation
- **Handles Non-linearity:** Captures complex relationships
- **Robust to Outliers:** Less sensitive than single trees

Confusion Matrix

Overview Detailed Analysis Model Evaluation Predictor Tool



Model Accuracy
99.9199919
991999%
Using Logistic Regression



Algorithm Used
Logistic
Regression
Classification Model



Training Samples
40001
Used for model training



Testing Samples
9999
Used for model testing

Confusion Matrix Heatmap

Confusion Matrix - Logistic Regression

Overall Accuracy: 99.9 %



Model Metrics

Metric	Value
Overall Accuracy	99.9%
Low Sensitivity	100%
Low Specificity	100%
Medium Sensitivity	99.9%
Medium Specificity	99.9%
High Sensitivity	99.9%
High Specificity	100%

1. Model Overview

- Algorithm Used:** Logistic Regression (a classification model)
- Training Samples:** 40,001
- Testing Samples:** 9,999
- Reported Model Accuracy:** 99.92% (very high)

3. Summary & Implications

- The model performs exceptionally well on this dataset, with near-perfect classification across all three classes.
- The confusion matrix shows minimal confusion between Medium and High classes (7 misclassifications total).
- Such high metrics suggest the task might be well-defined, the features highly predictive, or the classes well-separated.
- In real-world applications, accuracy this high could also prompt verification for potential data leakage, overfitting, or an overly simple task.

2. Confusion Matrix Heatmap

This visual shows how well the model predicted three classes: Low, Medium, and High.

- Rows** = Actual class
- Columns** = Predicted class
- The diagonal (top-left to bottom-right) shows correct predictions:
 - Low: 3,300 correct
 - Medium: 3,315 correct
 - High: 3,376 correct
- Very few errors:
 - 4 High instances misclassified as Medium
 - 3 Medium instances misclassified as High
 - 1 Low instance misclassified as Medium
 - 0 Low misclassified as High



Thanks